New Products

COMPONENTS

Ethernet PHYs

The LAN867x family of 10BASE-T1S PHYs, a new solution from Microchip Technology Inc., expands Ethernet connectivity to the very edges of industrial networks, simplifying archi-

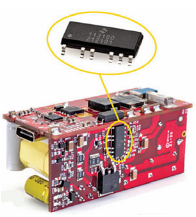
tectures and reducing risk for designers. The Ethernet physical layer (PHY) transceivers are high-performance, small-footprint devices enabling connections to standard system devices including sensors and actuators that previously required their own communication systems. Validated to the 10BASE-T1S standard for single-pair Ethernet released by IEEE, the device series addresses the challenges of creating all-Ethernet architectures for industrial applications such as process controls, building automation and consolidation of systems with multiple interconnection technologies.

Microchip Technology

www.microchip.com

Active clamp flyback controller

Silanna Semiconductor has launched a new integrated active clamp flyback (ACF) controller that will reduce the operational and no-load/stand-by power consumption of high-efficiency chargers and adaptors while driving down com-



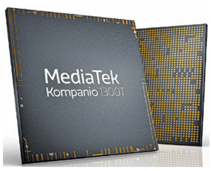
ponent count, bill-of-material cost and size. The SZ1131 ACF controller simplifies design and improves performance while addressing environmental sustainability goals through more efficient energy use. In the case of the SZ1131, efficiencies of up to 95% across universal (90Vac – 265Vac) input voltages and varying loads, combined with ultra-low no-load power of less than 20mW (including USB-PD applications) significantly reduce overall energy consumption.

Silanna Semiconductor

www.silanna.com

Powerful computing SoC

MediaTek has launched the Kompanio 1300T, an SoC designed to power incredible experiences across comput-



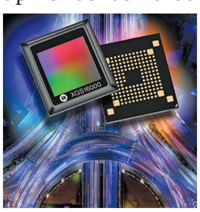
ing devices. The chip brings premium performance to tablets with advanced 5G, multimedia, AI, and gaming technologies for a premium user experience. The SoC allows OEMs to build powerful, lightweight, and portable tablets for online education, business, streaming services, gaming, and AI applications. The Kompanio 1300T chip is built on TSMC's 6nm process technology and integrates an octa-core CPU with high-performance Arm Cortex-A78 cores and power-efficient Arm Cortex-A55 cores. In addition, its high performance 9-core Arm Mali-G77 MC9 GPU provides premium computing performance and supports high frame rates for longer, smoother gaming.

MediaTek

www.mediatek.com

Image sensor

onsemi has introduced the XGS 16000 CMOS image sensor, which is a 16-megapixel sensor that provides high qual-



ity, global shutter imaging for factory automation applications including robotics and inspection systems. Consuming only one watt at 65 frames per second, it delivers exceptional performance at low power. This makes the XGS 16000 one of the best in class for power consumption, while also offering one of the highest resolutions available for standard 29mm x 29mm industrial cameras. It is designed in a unique 1:1 square aspect ratio, which helps maximise the image capture area within the optical circle of the camera lens and ensures optimal light sensitivity.

onsemi

www.onsemi.com

AI solution

Renesas Electronics and Syntiant Corp. (a deep learning chip technology company) have jointly developed a voice-controlled multimodal AI solution

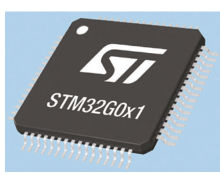
that enables low-power contactless operation for image processing in vision AI-based IoT and edge systems such as self-checkout machines, security cameras, and video conference systems, and smart appliances such as robotic cleaning devices. The new solution combines the Renesas RZ/V series vision AI microprocessor unit and the low-power multimodal, multi-feature Syntiant NDP120 neural decision processor to deliver advanced voice and image processing capabilities.

Renesas Electronics

www.renesas.com

Microcontroller series

STMicroelectronics has expanded its STM32G0 Arm Cortex-M0+ microcontroller (MCU) series with more product variants and features such as dual-bank Flash, support for CAN FD, and crystal-less USB full-speed data/host



support. The new STM32G050 Value Line and mainstream STM32G051 and STM32G061